

Lesson 09: Special Second-Order ODEs - Cauchy-Euler & Nonlinear

Professor Dave Explains Differential Equations Master Series · Lesson 9

Video: Special Second-Order Differential Equations: Cauchy-Euler, Nonlinear, and More **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that when coefficients are not constant but scale powers of x alongside derivatives ($ax^2y'' + bxy' + cy = 0$), the equation possesses a beautiful dilation invariance called the Cauchy-Euler equidimensional form. Instead of the exponential ansatz e^{rx} , we test power functions $y = x^m$. Additionally, Dave covers reduction of order: when an equation is missing the dependent variable y , substitute $v = y'$ to drop the order from two to one!

2. Core Theoretical Framework & Rigorous Formulations

1. Cauchy-Euler Equidimensional Equations

$$ax^2 \frac{d^2 y}{dx^2} + bx \frac{dy}{dx} + cy = 0 \quad (x > 0)$$

Testing $y = x^m \implies y' = mx^{m-1}, y'' = m(m-1)x^{m-2}$ produces the **Cauchy-Euler Auxiliary Equation**:

$$am(m-1) + bm + c = am^2 + (b-a)m + c = 0$$

- **Distinct Real Roots** $m_1 \neq m_2$: $y(x) = c_1 x^{m_1} + c_2 x^{m_2}$
- **Repeated Real Root** $m_1 = m_2 = m$: $y(x) = c_1 x^m + c_2 x^m \ln x$
- **Complex Conjugates** $\alpha \pm i\beta$: $y(x) = x^\alpha (c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x))$

2. Reduction of Order for Missing Dependent Variable

If $F(x, y', y'') = 0$ (no explicit y), substitute $v = y' \implies v' = y''$. This reduces the ODE to a 1st-order equation in $v(x)$.

3. Fully Solved Whiteboard Problems

Concordia Exam Classic: Cauchy-Euler Auxiliary Equation Trap

Problem Statement: Solve the Cauchy-Euler differential equation $x^2 y'' + 5xy' + 4y = 0$ for $x > 0$.

Step 1: Derive Auxiliary Equation with $m(m-1)$ Factor

$$1 \cdot m(m-1) + 5m + 4 = 0 \implies m^2 - m + 5m + 4 = 0 \implies m^2 + 4m + 4 = 0.$$

Step 2: Factor and Identify Root Multiplicity

$$(m+2)^2 = 0 \implies m_1 = m_2 = -2 \text{ (repeated real root).}$$

Step 3: Construct Linearly Independent Fundamental Solution Set

First solution: $y_1(x) = x^{-2}$. Second solution by reduction of order / dilation symmetry: $y_2(x) = x^{-2} \ln x$.

Step 4: Assemble General Solution

$$y(x) = c_1 x^{-2} + c_2 x^{-2} \ln x = \frac{c_1 + c_2 \ln x}{x^2} \text{ for } x > 0.$$

⚠ Common Exam Pitfall & Trapdoor Avoidance

Dropping the $-m$ term from $m(m-1)$ (writing $m^2 + 5m + 4 = 0$ instead of $m^2 + 4m + 4 = 0$) is the single most pervasive Cauchy-Euler mistake in ENGR 213 exams!

4. Key Takeaways & Exam Strategy Formula Card

🎯 Core Takeaway & Exam Rule

Always write $am(m-1) + bm + c = 0$. In repeated roots, multiply by $\ln x$, not by x .

- Video reference: [Special Second-Order Differential Equations: Cauchy-Euler, Nonlinear, and More](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.